PowerPoint Presentation: Chapter 10 - Pneumococcal Vaccination Types & Recommendations

The Core Science of Pneumococcal Immunity

Slide 1: Re-cap & Chapter 10 Focus

Previous Session: Chapter 9

- Global pneumonia burden & vaccine-preventable deaths
- PCV13 & PPSV23 vaccine types overview
- Influenza vaccination seasons & strains
- Implementation strategies

Today's Focus: Chapter 10

- Deep dive into pneumococcal vaccine science
- Age-specific recommendations & special populations
- Immunogenicity & safety profiles
- Clinical decision-making frameworks

Slide 2: Streptococcus pneumoniae - The Pathogen

Microbiology Essentials:

- **Gram-positive diplococcus**
- **90+ serotypes** identified
- **Capsular polysaccharide** virulence factor
- **Nasopharyngeal carriage** in 10-40% healthy adults
- **Invasive potential:** Pneumonia, meningitis, sepsis

High-resolution SEM image of S. pneumoniae

Slide 3: Serotype Distribution - Indian Context

Most Common Serotypes in India (IPD cases):

1. **19F** (12-15%)
2. **14** (10-12%)
3. **6B/6A** (10-12%)
4. **1** (8-10%)
5. **19A** (6-8%)
6. **23F** (5-7%)

Note: Serotype distribution varies by age and geography *Serotype pie chart with Indian data*

Slide 4: Pneumococcal Vaccine Evolution

Timeline of Vaccine Development:

- **1977:** 14-valent polysaccharide vaccine
- **1983:** 23-valent polysaccharide vaccine (PPSV23)
- **2000:** 7-valent conjugate vaccine (PCV7)
- **2010:** 13-valent conjugate vaccine (PCV13)
- **Future:** 15-20 valent conjugates, universal vaccines

Timeline visualization with impact graphs

Slide 5: Vaccine Mechanism - T-cell Dependent vs Independent

Conjugate Vaccines (PCV13/PCV15):

- **T-cell dependent** immune response
- **Memory B-cells** generation
- **Avidity maturation** with booster doses
- **Mucosal immunity** reduces nasopharyngeal carriage
- **Herd protection** through reduced transmission

Polysaccharide Vaccines (PPSV23):

- **T-cell independent** antibody response
- **No immunological memory** generation
- **Hyporesponsive** to repeat vaccinations
- **Short-lived immunity** (3-5 years)
- **No herd protection**

Comparison table with immunologic pathways

Slide 6: PCV13 Immunogenicity Data

Antibody Response Metrics:

- **Geometric Mean Titer (GMT):** 4-8 fold higher than PPSV23
- **Opsonophagocytic Activity (OPA):** Functional antibody measurement
- **Memory B-cell Response:** Persistent immunity
- **Aging Adult Response:** Better than PPSV23 in elderly

Immunogenicity comparison graphs

Slide 7: PCV13 vs PPSV23 Efficacy Comparison

Outcome	PCV13 Efficacy	PPSV23 Efficacy	Statistical Notes
IPD Prevention	75-85%	50-60%	$p<0.01$ all serotypes

Outcome	PCV13 Efficacy	PPSV23 Efficacy	Statistical Notes
Pneumonia Hospitalization	40-50%	20-30%	$p<0.05$ in elderly
Mortality Reduction	30-40%	15-25%	$p<0.01$ meta-analysis
Herd Immunity	35-50%	0-5%	$p<0.001$ in children

Head-to-head comparison with confidence intervals

Slide 8: Age-Specific Schedule - Infants <2 Years

Standard Immunization Schedule:

Age	Vaccine	Route	Dose	Rationale
6 weeks	PCV13-1	IM	0.5 mL	Early protection start
10 weeks	PCV13-2	IM	0.5 mL	Prime immune response
14 weeks	PCV13-3	IM	0.5 mL	Boost antibody levels
9-12 months	PCV13-4	IM	0.5 mL	Prolonged immunity

Total: 4 doses for optimal protection *Schedule timeline with maternal antibody decline overlay*

Slide 9: Catch-up Vaccination - Children

Unvaccinated or Partially Vaccinated:

Age	Recommended Schedule
7-11 months	3 doses (4-week intervals)
12-23 months	2 doses (8-week interval)
24-59 months	1 dose
≥60 months	Not routinely needed

Note: Complete series regardless of previous vaccines *Decision tree for catch-up schedules*

Slide 10: Immunocompromised Children

High-risk groups requiring vaccination:

- **HIV-infected children** (CD4 count consideration)
- **Malignancy patients** (timing with chemotherapy)
- **Solid organ transplant** recipients
- **Congenital immunodeficiencies**
- **Nephrotic syndrome** patients

Special schedules: Additional doses may be needed *High-risk categories with specific recommendations*

Slide 11: Adult Vaccination Strategy

Sequential Approach for Comprehensive Protection:

1. **PCV13 First** - Broad serotype coverage with memory response
2. **PPSV23 Later** (8+ weeks) - Additional serotypes, booster effect
3. **Monitoring Response** - Antibody titers if high-risk

Why Sequential:

- PCV13 enhances PPSV23 immunogenicity
- Broader protection against all common serotypes
- Longer duration of protection

Sequential vaccination flowchart

Slide 12: Elderly Adults (≥65 years)

Universal Recommendation:

- **PCV13 first dose** (any time after 65)
- **PPSV23 eight weeks later**
- **No routine re-vaccination** unless high-risk
- **Shared clinical decision-making**

Effectiveness Data:

- **75% reduction** in invasive pneumococcal disease
- **45% reduction** in pneumonia hospitalization
- **NNT = 1,200** to prevent one IPD case annually

Population-level impact statistics

Slide 13: High-Risk Adults <65 Years

Chronic Diseases:

- **COPD and asthma:** 2-4 fold increased risk
- **Diabetes mellitus:** Impaired immune response
- **Congestive heart failure:** Compromised respiratory reserve
- **Chronic liver disease:** Reduced opsonization

Immunosuppressive States:

- **HIV infection:** CD4 count-guided decisions
- **Malignancy:** Timing with chemotherapy cycles
- **Solid organ transplant:** 3-6 months post-transplant

- **Hematopoietic stem cell transplant:** +3 months post-transplant

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Slide 14: Special Populations - Asplenic Patients

Lifelong Risk of Overwhelming Post-Splenectomy Infection (OPSI):

- **Immediate PCV13** post-splenectomy (if not received)
- **PPSV23 8+ weeks later**
- **Re-vaccination every 5 years**
- **Lifelong antibiotic prophylaxis**
- **Medical alert bracelet** essential

Immunology: Spleen critical for pneumococcal clearance *OPSI pathophysiology and prevention strategy*

Slide 15: Pregnancy & Maternal Immunization

Safety Evidence:

- **PCV13 safe** in all trimesters (inactivated vaccine)
- **No adverse pregnancy outcomes** in large cohort studies
- **Placental antibody transfer** provides neonatal protection
- **Optimal timing:** Second/third trimester

Benefits:

- **Maternal protection** against invasive disease
- **Neonatal antibodies** from placental transfer
- **Reduced vertical transmission** risk
- **Postpartum vaccination** coverage

Slide 16: Healthcare Workers - Occupational Health

Ethical Obligation to Protect Patients:

- **Personal vaccination** to prevent nosocomial transmission
- **Annual influenza + pneumococcal** as appropriate
- **Serological testing** if occupational exposure
- **Infection control** integration

Influenza Protection Priority:

- **High-dose vaccine** if ≥65 years
- **Quadrivalent coverage** for comprehensive protection
- **Staff vaccination clinics** for convenience
- **Mandatory reporting** of influenza-like illness

Slide 17: Vaccine Safety Profile

PCV13 Safety (Millions of doses administered):

- **Injection site reactions:** Pain, swelling (10-30%)
- **Systemic reactions:** Fever, irritability (<2%)
- **Serious adverse events:** Extremely rare (<1:1 million)
- **No causal association** with any chronic conditions

PPSV23 Safety:

- **Similar reaction rates** to PCV13
- **No anaphylaxis** reported in trials
- **Safe in elderly** with multiple comorbidities
- **No drug interactions**

Slide 18: Adverse Events Following Immunization (AEFI)

Local Reactions:

- Tenderness, redness, swelling at injection site
- Usually mild and self-limiting (24-48 hours)
- Ice application and acetaminophen for comfort

Systemic Reactions:

- Fever, myalgia, fatigue (uncommon)
- Usually within first 24-72 hours
- Self-limiting, supportive care only

Rare but Serious Events:

- Allergic reactions (1:1 million doses)
- Cellulitis at injection site (<1:100,000)
- Neurological events (under investigation)

Slide 19: Contraindications & Precautions

Absolute Contraindications:

- **Severe allergic reaction** to any vaccine component
- **Anaphylaxis** to previous pneumococcal vaccine
- **Not applicable** for minor egg allergies

Precautions:

- **Moderate/severe illness:** Defer until recovery
- **Pregnancy:** Category B - generally safe
- **Immunosuppression:** May have reduced response
- **Breastfeeding:** Safe, compatible with lactation

Slide 10: Co-administration with Other Vaccines

Compatibility Data:

- **Safe with influenza vaccines** (simultaneous administration)
- **Compatible with childhood vaccines** (pediatric schedules)
- **No interference** with immune response
- **Preferred approach** for better compliance

Spacing Guidelines:

- **DTaP/IPV/Hib/HepB:** Can be given together
- **Measles-Mumps-Rubella:** 4-week separation
- **Live vs Inactivated:** No minimum interval required
- **Documentation:** Record all vaccines on same date

Slide 11: Immunogenicity in Special Populations

HIV-Infected Individuals:

- PCV13 safe regardless of CD4 count
- Response may be reduced (<200 cells/μL)
- Additional doses may be beneficial
- Timing: Any CD4 count >100 cells/μL

Cancer Patients:

- Timing crucial: Pre-chemotherapy when possible
- Reduced response during active treatment
- May need higher doses or additional boosters
- Coordinate with oncology treatment plan

Slide 12: Vaccine-Induced Serotype Replacement

Phenomemon Explanation:

- **Capsule switching** by pneumococci under vaccine pressure
- **Emergence of non-vaccine serotypes** (especially 19A, 35B)
- **Partial offset** of vaccine benefits
- **Under active surveillance**

Real-world Impact:

- **Net positive effect** despite replacement
- **Broad coverage vaccines** (PCV15, PCV20) in development
- **Ongoing monitoring** essential

Slide 13: Implementation Challenges in India

Access Barriers:

- **Geographic distribution** in rural/remote areas
- **Supply chain logistics** for cold chain maintenance
- **Healthcare worker training** in proper administration
- **Community awareness** and demand generation

System-level Issues:

- **Program coordination** between public/private sectors
- **Data management** for coverage monitoring
- **Resource allocation** for vaccination campaigns
- **Quality assurance** across different delivery platforms

Slide 14: Cost-Effectiveness in Indian Context

Economic Analysis:

- **Cost per vaccinated child:** ₹500-800 including service delivery
- **DALYs averted:** 8-12 per vaccinated child
- **ROI:** 8-12 times investment recovered through reduced healthcare costs
- **Long-term savings:** Reduced antibiotic resistance burden

Bottleneck Analysis:

- **Vaccine pricing** vs local manufacturing capacity
- **Delivery costs** in difficult terrain
- **Training expenses** for healthcare workforce
- **Monitoring costs** for adverse event surveillance

Slide 15: Future Pneumococcal Vaccines

Next-Generation Vaccines:

PCV15 & PCV20:

- **Broader serotype coverage** (15 and 20 serotypes)
- **Same conjugate technology** with enhanced immunogenicity
- **Backward compatibility** with PCV13
- **Global rollout** ongoing

Universal Pneumococcal Vaccines:

- **Conserved antigens** approach (PhtD, Ply)
- **Serotype-independent** protection
- **Potential for single-dose** immunity
- **Elimination goal** attainment

Protein-Based Vaccines:

- **Pneumococcal surface protein A** (PspA)
- **Pneumolysin** toxoid
- **Combination approaches** for comprehensive protection

Slide 16: CBME Assessment - Clinical Decision Making

Case Scenario: *68-year-old male with COPD, recently hospitalized for pneumonia. Unvaccinated for pneumococcus. What would you recommend?*

Options Analysis:

1. **PPSV23 only:** Adequate but suboptimal coverage
2. **PCV13 only:** Good choice but incomplete protection
3. **PCV13 → PPSV23: Optimal approach ★**
4. **Both simultaneously:** Increased side effects risk

Slide 17: Key Clinical Pearls

Pneumococcal Vaccination Wisdom:

1. **Start with PCV13** for memory response and broadest protection
2. **Add PPSV23 later** for additional serotypes and booster effect
3. **Age ≥65 = automatic indication** for vaccination
4. **High-risk conditions** warrant vaccination regardless of age
5. **Pregnancy safe** with PCV13 in all trimesters
6. **Asplenic patients** need lifelong protection strategy
7. **Herd immunity** benefits unvaccinated populations
8. **Cost saves lives** - every vaccine dose prevents suffering

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Slide 18: Interactive Quiz

Test your pneumococcal knowledge:

Question: Best pneumococcal vaccine for 70-year-old diabetic?

- A) PPSV23 only
- B) PCV13 only
- C) **PCV13 then PPSV23** ★
- D) Neither routinely needed

Question: PCV13 covers how many pneumococcal serotypes?

- A) 7 serotypes
- B) **13 serotypes** ★
- C) 23 serotypes
- D) All serotypes

Slide 19: References & Further Reading

📚 **Recommended Resources:**

1. **ACIP Adult Immunization Guidelines (2024)**
2. **WHO Pneumococcal Vaccine Position Paper (2023)**
3. **CDC Pink Book: Pneumococcal Disease (2024)**
4. **Indian Academy of Pediatrics Guidelines (2022)**
5. **Vaccine Journal - PCV13 immunogenicity studies**

Slide 20: Session Wrap-up

What We Learned Today:

- **Immunologic basis** for different vaccine mechanisms
- **Evidence-based schedules** for all age groups
- **Risk-benefit profiles** for safe vaccination practice
- **Implementation challenges** and solutions
- **Future vaccine prospects** for complete pneumococcal control

Tomorrow: Chapter 11 - Influenza Vaccination Strategies

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